

Mobile App Tasks — Offline Shop Management App

Epic: MOB-EPIC-1 — Build Mobile App for Offline Scanning and Bluetooth Sync

Purpose

Build the mobile app used for offline barcode scanning, stock counting, stock addition, stock removal, physical inventory count, unknown product handling, and Bluetooth sync with the Qt app.

Component / Area

Mobile App

Dependency

Qt app sync protocol
Mobile local database
Bluetooth connection
Barcode scanner / camera access

Phase 1 — Mobile App Foundation

MOB-1 — Create Mobile App Project Skeleton

Purpose

Create the base mobile app project structure.

Component / Area

Mobile App / Foundation

Dependency

None

Detailed Description

Create the initial mobile app project with organized folders for screens, services, local database, Bluetooth, barcode scanning, sync, models, and shared utilities.

The app should support future screens such as login, dashboard, add stock, remove stock, physical count, unknown list, product info entry, pending sync, and Bluetooth connection.

Acceptance Criteria

Mobile app project is created.

App can start successfully.

Base navigation exists.

Folder structure is organized.

Placeholder screens can be opened.

MOB-2 — Implement Mobile App Configuration

Purpose

Load required mobile app configuration.

Component / Area

Mobile App / Configuration

Dependency

MOB-1

Detailed Description

Implement configuration for device ID, Bluetooth settings, local database settings, QR token rules, and sync behavior.

Acceptance Criteria

Mobile app can load configuration.

Device ID is available to sync and transaction records.
Bluetooth settings are available.
Invalid config shows clear error.

MOB-3 — Implement Local Database Layer

Purpose

Allow the mobile app to work offline.

Component / Area

Mobile App / Local Database

Dependency

MOB-1

Detailed Description

Implement the local database used by the mobile app.

The mobile app should store local user data, product copy, unknown products, pending stock additions, pending stock removals, physical count sessions, product info suggestions, and sync batch status.

Acceptance Criteria

Local database is created.

Database can be opened successfully.

Database errors are handled.

Tables or local storage models exist for required mobile data.

UI code does not directly access database queries.

MOB-4 — Implement Local Repository Layer

Purpose

Create reusable local data access classes.

Component / Area

Mobile App / Local Repository

Dependency

MOB-3

Detailed Description

Create repository classes for local users, local products, unknown products, pending stock additions, pending stock removals, physical count sessions, product info suggestions, and sync batches.

Acceptance Criteria

LocalUserRepository exists.

LocalProductRepository exists.

UnknownProductRepository exists.

PendingStockAdditionRepository exists.

PendingStockRemovalRepository exists.

PhysicalCountRepository exists.

ProductInfoSuggestionRepository exists.

SyncBatchRepository exists.

Repositories do not contain UI logic.

Phase 2 — Mobile Login and Session

MOB-5 — Implement Mobile Login Page

Purpose

Allow users to log in on the mobile app.

Component / Area

Mobile App / Authentication UI

Dependency

MOB-1

Detailed Description

Create the login page with user ID/password login and QR code login option.

Acceptance Criteria

Login page has user_id input.

Login page has password input.

Login button exists.

QR login button exists.

Empty fields show validation error.

Successful login opens mobile dashboard.

MOB-6 — Implement Offline Password Login

Purpose

Allow login using synced local user data.

Component / Area

Mobile App / Authentication Logic

Dependency

MOB-4, MOB-5

Detailed Description

Implement login verification using local synced user data.

The mobile app should not connect to PostgreSQL.

Inactive users should not be allowed to log in.

Acceptance Criteria

Mobile login uses local user database.

Valid active user can log in.

Invalid password is rejected.

Unknown user is rejected.

Inactive user is rejected.

Plain password is not stored.

MOB-7 — Implement QR Code Login

Purpose

Allow faster mobile login using QR code.

Component / Area

Mobile App / QR Login

Dependency

MOB-5, MOB-6

Detailed Description

Implement QR scan login using `user_id`, `login_token`, and `expires_at`.

QR code must not contain password, password hash, birthday, or mobile number.

Acceptance Criteria

QR login screen opens camera scanner.

QR content is parsed.

Expired token is rejected.

Invalid token is rejected.

Inactive user is rejected.

Valid token logs in user.

QR does not require internet.

MOB-8 — Implement Current User Session

Purpose

Track the logged-in user during mobile app usage.

Component / Area

Mobile App / Session

Dependency

MOB-6, MOB-7

Detailed Description

Create a current user session model with user ID, name, role, login method, and login time.

This session should be used when saving pending sync data and operation metadata.

Acceptance Criteria

Current user session is created after login.

Session stores user_id.

Session stores user name.

Session stores role.

Session stores login method.

Session is available to all operation screens.

Logout clears session.

MOB-9 — Implement Mobile Dashboard

Purpose

Show main mobile operation menu.

Component / Area

Mobile App / Dashboard

Dependency

MOB-8

Detailed Description

Create the mobile dashboard with navigation to all mobile operation screens.

Acceptance Criteria

Dashboard opens after login.

User can open Add Stock.

User can open Remove Stock.

User can open Physical Inventory Count.

User can open Unknown List.

User can open Product Info Entry.

User can open Pending Product Info Suggestions.

User can open Pending Sync.

User can open Bluetooth Connection.
User can open Current User.
User can logout.

Phase 3 — Bluetooth Connection

MOB-10 — Implement Bluetooth Connection Page

Purpose

Allow the mobile app to connect to the Qt app.

Component / Area

Mobile App / Bluetooth UI

Dependency

MOB-2

Detailed Description

Create Bluetooth connection UI showing connected/disconnected status, paired Qt device, last sync time, and pending sync count.

Acceptance Criteria

Bluetooth Connection page exists.
Connection status is displayed.
Paired Qt device is displayed if available.
User can attempt connection.
Successful connection is shown.
Connection failure is shown.
Last sync time is displayed.
Pending sync count is displayed.

MOB-11 — Implement Bluetooth Send Message Service

Purpose

Send sync batches and scan events to Qt.

Component / Area

Mobile App / Bluetooth

Dependency

MOB-10

Detailed Description

Implement Bluetooth message sending service.

The service should support sending unknown barcode batches, stock addition batches, stock removal batches, physical count sessions, and product info suggestions.

Acceptance Criteria

Mobile can send Bluetooth message to Qt.

Message contains type.

Message contains batch_id when needed.

Send failure is handled.

Bluetooth service is separated from UI pages.

MOB-12 — Implement Bluetooth Receive Message Service

Purpose

Receive ACK and data from Qt.

Component / Area

Mobile App / Bluetooth

Dependency

MOB-10, MOB-11

Detailed Description

Implement Bluetooth receive handling for ACK messages, user sync data, product sync data, and product info suggestion review results.

Acceptance Criteria

Mobile can receive Bluetooth messages from Qt.

ACK messages are parsed.

User sync messages are parsed.

Product sync messages are parsed.

Product suggestion review results are parsed.

Invalid messages are rejected safely.

Phase 4 — Incoming Sync from Qt

MOB-13 — Implement User Data Sync Receiver

Purpose

Receive official user data from Qt.

Component / Area

Mobile App / Sync / Users

Dependency

MOB-12, MOB-4

Detailed Description

Implement receiver for user data sent by Qt.

Mobile should save mobile-safe login data locally.

Plain passwords must never be received or stored.

Acceptance Criteria

Mobile receives user sync batch.

User records are saved locally.

User role is saved.

User status is saved.
Login verifier/hash is saved.
Plain password is not stored.
Mobile sends ACK after successful save.

MOB-14 — Implement Product Info Sync Receiver

Purpose

Receive official product data from Qt.

Component / Area

Mobile App / Sync / Products

Dependency

MOB-12, MOB-4

Detailed Description

Implement receiver for product info sent by Qt.

The mobile app should update its local product database.

If product info resolves an unknown barcode, mobile should remove that unknown item and delete its local picture.

Acceptance Criteria

Mobile receives product info sync batch.
Product records are saved locally.
Existing product records are updated.
Resolved unknown barcode is removed from Unknown List.
Local picture for resolved unknown barcode is deleted.
Mobile sends ACK after successful save.

MOB-15 — Implement Product Suggestion Review Result Receiver

Purpose

Receive review results for product info suggestions from Qt.

Component / Area

Mobile App / Sync / Product Suggestions

Dependency

MOB-12, MOB-4

Detailed Description

Implement receiver for accepted, edited-and-accepted, or rejected product info suggestion results.

If accepted, update local product data.

If rejected, mark the suggestion as rejected.

Acceptance Criteria

Mobile receives accepted suggestion result.

Mobile receives edited-and-accepted suggestion result.

Mobile receives rejected suggestion result.

Accepted product info updates local product database.

Rejected suggestion is marked rejected.

Mobile sends ACK after successful save.

Phase 5 — Barcode Scanning

MOB-16 — Implement Barcode Scanner Component

Purpose

Allow users to scan product barcodes.

Component / Area

Mobile App / Barcode Scanner

Dependency

MOB-1

Detailed Description

Implement reusable barcode scanning component used by Add Stock, Remove Stock, Physical Count, Unknown List, and Product Info Entry.

Acceptance Criteria

Barcode scanner opens camera.

Barcode can be detected.

Scanned barcode is returned to caller screen.

Scan failure is handled.

Manual barcode input is available as fallback.

MOB-17 — Implement Local Product Lookup

Purpose

Find product information after scanning barcode.

Component / Area

Mobile App / Product Lookup

Dependency

MOB-4, MOB-16

Detailed Description

Implement local product lookup by barcode.

The lookup should use the local product database synced from Qt.

Acceptance Criteria

Scanned barcode can be searched locally.

Known product returns product info.

Unknown product returns not found.

Lookup works offline.

Phase 6 — Unknown Product Handling

MOB-18 — Implement Unknown List Page

Purpose

Show unknown products stored on mobile.

Component / Area

Mobile App / Unknown Products

Dependency

MOB-4, MOB-16

Detailed Description

Create Unknown List screen showing unknown barcodes, local pictures, temporary info, pending quantities, and sync status.

Pictures stay only on mobile and are not sent to Qt.

Acceptance Criteria

- Unknown List page exists.
- Unknown products are displayed.
- Each unknown item shows barcode.
- Each unknown item can show local picture.
- Temporary name is displayed if available.
- Temporary price is displayed if available.
- Pending quantity is displayed if available.
- Sync status is displayed.

MOB-19 — Implement Add Unknown Barcode Flow

Purpose

Create or update unknown product records.

Component / Area

Mobile App / Unknown Products

Dependency

MOB-17, MOB-18

Detailed Description

When a scanned barcode is not found locally, mobile should add it to Unknown List.

If the barcode already exists in Unknown List, update last_seen_at and quantity instead of creating duplicate records.

Acceptance Criteria

Unknown barcode can be added.

Duplicate unknown barcode is not created.

Existing unknown barcode updates last_seen_at.

Pending quantity can be updated.

created_by_user_id is saved.

device_id is saved.

MOB-20 — Implement Local Picture for Unknown Products

Purpose

Allow local visual reference for unknown products.

Component / Area

Mobile App / Unknown Products / Local Picture

Dependency

MOB-18, MOB-19

Detailed Description

Allow the user to take, replace, or delete a picture for an unknown barcode.

The picture must stay only on mobile.

It must not be included in sync payloads to Qt.

Acceptance Criteria

User can take picture for unknown barcode.

User can replace picture.

User can delete picture.

Only one representative picture is kept per unknown barcode.

Picture is stored locally only.

Picture is not sent to Qt.

MOB-21 — Implement Unknown List Sync Payload

Purpose

Prepare unknown barcode data for Qt sync.

Component / Area

Mobile App / Sync / Unknown Products

Dependency

MOB-18, MOB-19, MOB-20

Detailed Description

Build sync payload for Unknown List.

Payload should include barcode and metadata only.

It should not include product picture.

Acceptance Criteria

Unknown List sync payload includes batch_id.

Payload includes barcode.

Payload includes temporary product name if available.

Payload includes temporary price if available.

Payload includes pending quantity if available.

Payload includes first_seen_at.

Payload includes last_seen_at.

Payload includes created_by_user_id.
Payload includes device_id.
Payload does not include image data.

MOB-22 — Implement Send Unknown List to Qt

Purpose

Sync unknown barcode list to Qt.

Component / Area

Mobile App / Sync / Unknown Products

Dependency

MOB-11, MOB-21

Detailed Description

Send Unknown List batch to Qt when Bluetooth is available.

After Qt ACK, mark unknown barcodes as sent_to_qt but do not delete them yet.

Delete unknown item and local picture only after official product info is received from Qt.

Acceptance Criteria

User can send Unknown List to Qt.

Bluetooth disconnected shows clear error.

Batch is saved before sending.

ACK marks unknown items as sent_to_qt.

No picture is sent.

Unknown item is not deleted after ACK alone.

Unknown item is deleted only after product info sync from Qt.

Phase 7 — Add Stock

MOB-23 — Implement Add Stock Page

Purpose

Allow users to add products to stock.

Component / Area

Mobile App / Stock Addition

Dependency

MOB-16, MOB-17, MOB-19

Detailed Description

Create Add Stock page where user scans barcode, sees product info if known, enters quantity, stock-entry price, source information, supplier, carried-by, and note.

Acceptance Criteria

Add Stock page exists.
User can scan barcode.
Known product info is displayed.
Unknown barcode is added to Unknown List.
User can enter quantity.
User can enter stock-entry price.
User can enter source location.
User can enter supplier name.
User can enter carried_by.
User can enter note.

MOB-24 — Implement Repeated Quantity Entry for Add Stock

Purpose

Allow fast quantity entry.

Component / Area

Mobile App / Stock Addition

Dependency

MOB-23

Detailed Description

Allow the user to type quantity directly instead of scanning one by one.

Acceptance Criteria

User can enter quantity manually.

Quantity must be greater than zero.

Repeated scan increases quantity.

Manual quantity and scanned quantity are handled correctly.

MOB-25 — Implement Save Pending Stock Addition

Purpose

Save stock addition locally before Qt confirmation.

Component / Area

Mobile App / Stock Addition / Pending Sync

Dependency

MOB-23, MOB-24, MOB-4

Detailed Description

Save stock addition as pending sync when Bluetooth is unavailable or when live send fails.

The pending record should contain barcode, quantity, stock-entry price, source fields, user ID, device ID, and batch metadata.

Acceptance Criteria

Pending stock addition can be saved.

barcode is saved.

quantity is saved.

unit_price is saved.

source_location is saved if provided.

supplier_name is saved if provided.

carried_by is saved if provided.

created_by_user_id is saved.

device_id is saved.
sync_status is pending.

MOB-26 — Implement Send Stock Addition to Qt

Purpose

Send stock addition batch to Qt.

Component / Area

Mobile App / Sync / Stock Addition

Dependency

MOB-11, MOB-25

Detailed Description

Send pending stock addition data to Qt.

Mobile should mark it as synced only after Qt ACK.

Acceptance Criteria

Stock addition batch can be sent to Qt.

Batch contains unique batch_id.

Line items contain line_id.

ACK marks batch as synced.

No ACK keeps batch pending.

Send failure keeps batch pending.

Phase 8 — Remove Stock

MOB-27 — Implement Remove Stock Page

Purpose

Allow users to remove products from stock.

Component / Area

Mobile App / Stock Removal

Dependency

MOB-16, MOB-17, MOB-19

Detailed Description

Create Remove Stock page where user scans barcode, enters quantity, selects removal reason, and enters outgoing destination information.

Acceptance Criteria

Remove Stock page exists.

User can scan barcode.

Known product info is displayed.

Unknown barcode can be added to Unknown List.

User can enter removal quantity.

User can select removal reason.

User can enter destination type.

User can enter destination name.

User can enter destination contact.

User can enter destination note.

MOB-28 — Implement Repeated Quantity Entry for Remove Stock

Purpose

Allow fast stock removal quantity entry.

Component / Area

Mobile App / Stock Removal

Dependency

MOB-27

Detailed Description

Allow the user to enter removal quantity directly or increase quantity through repeated scans.

Acceptance Criteria

User can enter removal quantity manually.

Quantity must be greater than zero.

Repeated scan increases removal quantity.

Invalid quantity shows error.

MOB-29 — Implement Save Pending Stock Removal

Purpose

Save stock removal locally before Qt confirmation.

Component / Area

Mobile App / Stock Removal / Pending Sync

Dependency

MOB-27, MOB-28, MOB-4

Detailed Description

Save stock removal as pending sync when Bluetooth is unavailable or when live send fails.

The pending record should contain barcode, quantity, reason, destination fields, user ID, device ID, and batch metadata.

Acceptance Criteria

Pending stock removal can be saved.

barcode is saved.

quantity is saved.

removal reason is saved.

destination_type is saved if provided.

destination_name is saved if provided.

destination_contact is saved if provided.

destination_note is saved if provided.

created_by_user_id is saved.

device_id is saved.

sync_status is pending.

MOB-30 — Implement Send Stock Removal to Qt

Purpose

Send stock removal batch to Qt.

Component / Area

Mobile App / Sync / Stock Removal

Dependency

MOB-11, MOB-29

Detailed Description

Send pending stock removal data to Qt.

Mobile should mark it as synced only after Qt ACK.

Qt will make the final official stock decision.

Acceptance Criteria

Stock removal batch can be sent to Qt.

Batch contains unique batch_id.

Line items contain line_id.

Reason is included.

Destination fields are included if provided.

ACK marks batch as synced.

No ACK keeps batch pending.

Send failure keeps batch pending.

Phase 9 — Physical Inventory Count

MOB-31 — Implement Physical Inventory Count Page

Purpose

Allow users to count products physically existing in the shop.

Component / Area

Mobile App / Physical Count

Dependency

MOB-16, MOB-17, MOB-19

Detailed Description

Create Physical Inventory Count page where user starts a count session, scans products, and records counted quantities.

Physical count does not directly change stock.

Acceptance Criteria

Physical Inventory Count page exists.

User can start count session.

count_session_id is created.

User can scan barcode.

Known product count is recorded.

Unknown barcode is added to Unknown List.

Count session status is draft/counting.

MOB-32 — Implement Physical Count Quantity Entry

Purpose

Allow accurate counted quantity entry.

Component / Area

Mobile App / Physical Count

Dependency

MOB-31

Detailed Description

Allow repeated scans and manual quantity entry during physical count.

Acceptance Criteria

Repeated scan increases counted quantity.

User can enter counted quantity manually.

Quantity must be greater than or equal to zero.

Counted quantity is saved per barcode.

MOB-33 — Implement Save Pending Physical Count Session

Purpose

Save physical count locally before Qt review.

Component / Area

Mobile App / Physical Count / Pending Sync

Dependency

MOB-31, MOB-32, MOB-4

Detailed Description

Save the physical count session and count items locally.

The session should remain pending until synced to Qt.

Acceptance Criteria

Count session can be saved locally.

Session has count_session_id.

Session has started_by_user_id.

Session has started_at.

Session has ended_at if finished.

Session has device_id.

Count items are saved.

Unknown product items are marked.

Session sync_status is pending.

MOB-34 — Implement Send Physical Count Session to Qt

Purpose

Send physical inventory count session to Qt.

Component / Area

Mobile App / Sync / Physical Count

Dependency

MOB-11, MOB-33

Detailed Description

Send physical count session and items to Qt.

Mobile should mark the session as synced only after ACK.

Qt will compare counted quantity against official stock quantity.

Acceptance Criteria

Physical count batch can be sent to Qt.

Batch contains unique batch_id.

Session data is included.

Count items are included.

Unknown barcode items are included.

ACK marks session as synced.

No ACK keeps session pending.

Send failure keeps session pending.

Phase 10 — Mobile Product Info Entry and Suggestions

MOB-35 — Implement Product Info Entry Page

Purpose

Allow users to add or update product information on mobile.

Component / Area

Mobile App / Product Info Suggestions

Dependency

MOB-16, MOB-17, MOB-4

Detailed Description

Create Product Info Entry page where user can scan/select barcode and enter product fields.

Product info entered on mobile is not official. It is saved as a suggestion and sent to Qt for review.

Acceptance Criteria

Product Info Entry page exists.

User can scan barcode.

User can select known product.

User can enter product_name.

User can enter product_type.

User can enter country_of_manufacture.

User can enter default_price.

User can enter details.

Mobile saves data as suggestion, not official product info.

MOB-36 — Implement Create Product Info Suggestion

Purpose

Save new product info suggestion from mobile.

Component / Area

Mobile App / Product Info Suggestions

Dependency

MOB-35

Detailed Description

If the scanned barcode does not exist locally, mobile should create a product info suggestion with suggestion_type = create.

Acceptance Criteria

User can create suggestion for unknown barcode.
suggestion_type is create.
Required fields are validated.
Suggestion is saved locally.
Suggestion status is pending_sync.
created_by_user_id is saved.
device_id is saved.

MOB-37 — Implement Update Product Info Suggestion

Purpose

Save product update suggestion from mobile.

Component / Area

Mobile App / Product Info Suggestions

Dependency

MOB-35

Detailed Description

If the product exists locally, mobile should save changes as suggestion_type = update.

The local official product copy should not be overwritten until Qt accepts and sends official product info back.

Acceptance Criteria

User can create update suggestion for known product.
suggestion_type is update.

Original local product is not overwritten immediately.
Suggestion is saved locally.
Suggestion status is pending_sync.
created_by_user_id is saved.
device_id is saved.

MOB-38 — Implement Pending Product Info Suggestions Page

Purpose

Show product info suggestions waiting for sync or review.

Component / Area

Mobile App / Product Info Suggestions

Dependency

MOB-36, MOB-37

Detailed Description

Create page showing local suggestions and their statuses.

Acceptance Criteria

Pending Product Info Suggestions page exists.
Suggestions are listed.
Each item shows barcode.
Each item shows suggestion_type.
Each item shows status.
User can open suggestion detail.
User can delete local draft if not sent.
User can retry sync.

MOB-39 — Implement Send Product Info Suggestions to Qt

Purpose

Sync mobile product info suggestions to Qt.

Component / Area

Mobile App / Sync / Product Info Suggestions

Dependency

MOB-11, MOB-36, MOB-37, MOB-38

Detailed Description

Send create/update product info suggestions to Qt when Bluetooth is available.

Mobile should mark suggestions as sent_to_qt only after Qt ACK.

Acceptance Criteria

Suggestion batch can be sent to Qt.

Batch contains unique batch_id.

Suggestion type is included.

Suggested product fields are included.

ACK marks suggestion as sent_to_qt.

No ACK keeps suggestion pending.

Send failure keeps suggestion pending.

Phase 11 — Pending Sync

MOB-40 — Implement Pending Sync Page

Purpose

Show all mobile data waiting to sync with Qt.

Component / Area

Mobile App / Sync UI

Dependency

MOB-22, MOB-26, MOB-30, MOB-34, MOB-39

Detailed Description

Create Pending Sync page showing all pending unknown barcodes, stock additions, stock removals, physical count sessions, and product info suggestions.

Acceptance Criteria

Pending Sync page exists.
Pending unknown barcodes are shown.
Pending stock additions are shown.
Pending stock removals are shown.
Pending physical count sessions are shown.
Pending product info suggestions are shown.
User can sync individual batch.
User can sync all.

MOB-41 — Implement Sync All Flow

Purpose

Send all pending data to Qt.

Component / Area

Mobile App / Sync

Dependency

MOB-40, MOB-11, MOB-12

Detailed Description

Implement Sync All, sending pending batches one by one.

Each batch should wait for ACK before being marked as synced.

Acceptance Criteria

User can click Sync All.
Bluetooth disconnected shows error.
Pending batches are sent one by one.
Each batch waits for ACK.

Successful ACK marks batch synced.
Failed batch remains pending.
Sync result summary is shown.

MOB-42 — Implement Sync Batch State Management

Purpose

Track pending, sent, acknowledged, failed, and duplicate sync states.

Component / Area

Mobile App / Sync

Dependency

MOB-40, MOB-41

Detailed Description

Implement sync batch status updates.

Statuses should clearly show whether data is local only, sent to Qt, acknowledged, failed, or duplicate.

Acceptance Criteria

Sync batch can be pending.
Sync batch can be sending.
Sync batch can be acknowledged.
Sync batch can be failed.
Sync batch can be duplicate.
UI shows current sync status.
Failed sync can be retried.
Duplicate response is handled safely.

Phase 12 — Current User and QR Code

MOB-43 — Implement Current User Page

Purpose

Show current logged-in user information.

Component / Area

Mobile App / User Session UI

Dependency

MOB-8

Detailed Description

Create Current User page showing user ID, name, role, login method, and last login time.

Acceptance Criteria

Current User page exists.

Page shows user_id.

Page shows name.

Page shows role.

Page shows login method.

Page shows login time.

Logout button exists.

MOB-44 — Implement Generate QR Code Page

Purpose

Allow user to generate QR code for login.

Component / Area

Mobile App / QR Code

Dependency

MOB-43

Detailed Description

Generate QR code containing user_id, login_token, and expires_at.

The QR code must not contain password, password hash, birthday, or mobile number.

Acceptance Criteria

User can generate QR code.

QR contains user_id.

QR contains login_token.

QR contains expires_at.

QR does not contain password.

QR does not contain password hash.

QR does not contain birthday.

QR does not contain mobile number.

Expired token cannot be used.

Phase 13 — User-Friendly Messages and Safety Rules

MOB-45 — Implement User-Friendly Error Messages

Purpose

Show clear messages for common mobile problems.

Component / Area

Mobile App / Error Handling

Dependency

All mobile screens

Detailed Description

Implement consistent error and success messages across the mobile app.

Acceptance Criteria

Invalid login shows clear error.

Inactive user shows clear error.
Bluetooth disconnected shows clear error.
Sync failure shows clear error.
Invalid barcode shows clear error.
Invalid quantity shows clear error.
Unknown product message is clear.
Successful pending save shows success message.
Successful sync shows success message.

MOB-46 — Implement Offline-Safe Save Rule

Purpose

Prevent data loss when Bluetooth fails.

Component / Area

Mobile App / Data Safety

Dependency

MOB-25, MOB-29, MOB-33, MOB-39, MOB-41

Detailed Description

Ensure important data is saved locally before attempting Bluetooth sync.

The mobile app must never delete pending data before receiving ACK from Qt.

Acceptance Criteria

Stock additions are saved locally before sync.
Stock removals are saved locally before sync.
Physical count sessions are saved locally before sync.
Product info suggestions are saved locally before sync.
Unknown barcodes are saved locally before sync.
No pending data is deleted before ACK.
Failed sync keeps data pending.

MOB-47 — Implement Local Picture Cleanup Rule

Purpose

Delete unknown product pictures only after product info is confirmed from Qt.

Component / Area

Mobile App / Unknown Product Cleanup

Dependency

MOB-14, MOB-20

Detailed Description

When official product info is received from Qt for an unknown barcode, mobile should remove the unknown record and delete the local picture.

Acceptance Criteria

Resolved unknown barcode is detected from product info sync.

Unknown record is removed.

Local picture is deleted.

Pending temporary info is removed.

Known product info remains in local product database.

Picture is not deleted before Qt product info sync.

Recommended Build Order

1. MOB-1 to MOB-4
 2. MOB-5 to MOB-9
 3. MOB-10 to MOB-12
 4. MOB-13 to MOB-15
 5. MOB-16 to MOB-22
 6. MOB-23 to MOB-30
 7. MOB-31 to MOB-34
 8. MOB-35 to MOB-39
 9. MOB-40 to MOB-42
 10. MOB-43 to MOB-47
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MVP Cut Line

For the first working MVP, build these first:

MOB-1 to MOB-9
MOB-10 to MOB-14
MOB-16 to MOB-17
MOB-18 to MOB-22
MOB-23 to MOB-30
MOB-40 to MOB-42
MOB-45 to MOB-46

Then add:

Physical inventory count
Mobile product info suggestions
QR login
QR generation
Advanced current user page
Local picture cleanup polish